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# Communication Systems II

Constellation Design for Simultaneous Wireless Information and Power Transfer (SWIPT)

# 1 Task 1

## 1.1 Theoretical analysis about PAPR versus $d_{min}$ for 16-CQAM, 16-QAM, 16-PAM, 16-PSK

SWIPT is a new wireless technology which transmits energy and information wirelessly in the same resource. It seems to be a promising solution for massive machine type communications (mMTC), characterized by a vast number of low-rate low power devices, such as the Internet of Things (IoT).In this section, we compare a 16-Circular QAM with other known modulation schemes, namely 16-PAM, 16-PSK, and 16-QAM in terms of energy harvesting and signal error probability.
For achieving high levels of energy harvesting it's necessary to use signals with high peak to average power ratio (PAPR). However, there is a tradeoff between PAPR and minimum Euclidean distance ($d_{min}$). This means that having signals with high PAPR contributes to transferring energy however leads to high signal error probability (SEP), which prevents having effective information transfer.
At [1] the authors present a 16-CQAM which provides a flexible PAPR behavior versus minimum Euclidean distance, which allows us to tune the amount of energy to be harvested without affecting the information transfer performance. As we can see by simulation results at [1] CQAM can maintain a high PAPR as $d_{min}$ arises. When CQAM is compared to other modulation schemes, like 16-PSK and 16-PAM, it seems that it outperforms them. 16-QAM is getting very close to the performance of 16-CQAM and we can also observe that higher level CQAM achieve higher PAPR values but the error rate performance decreases rapidly.
So for the above-mentioned reasons we conclude that CQAM performs better than the other known modulation schemes as it provides flexible tradeoff between PAPR and $d_{min}$ that is important for SWIPT applications.

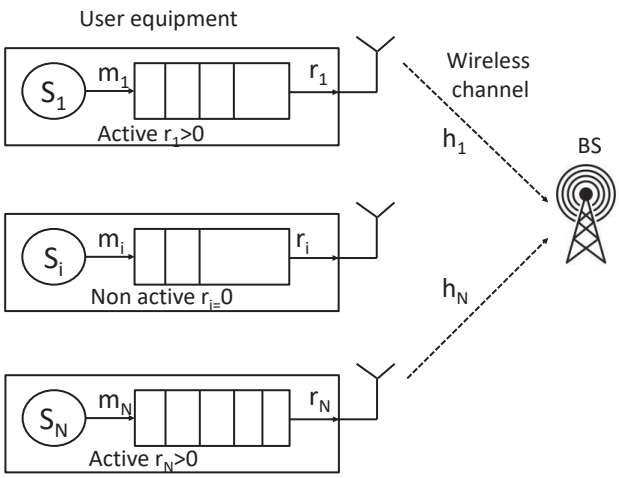


Figure 1: System Model

## 1.2 Simulation analysis

From [1] we have that PAPR for CQAM can be written as

$$PAPR = \frac{R_N^2}{E_i[R_i^2]}. \tag{1}$$

We have also that

$$R_1 = \frac{d_{min}}{2sin(\frac{\pi}{n})} \tag{2}$$

where n is the symbols per circle.
So we can write PAPR as a function of $d_{min}$. But first we might write all the radii from $R_2$ up to $R_{N-1}$ as a function of $R_1$ and then we 'll find $R_N$ using the condition

$$E_i[R_i] = 1 \tag{3}$$

Based on [1] we build the 16-CQAM in order to achieve better energy harvesting performance. According to the algorithm that they use, we have for 16-CQAM with 4 circles

$$R_1 = \frac{d_{min}}{\sqrt{2}} \tag{4}$$

For $R_2$

$$2d_{min} = \sqrt{R_2^2 + R_2^2}$$

$$d_{min} = \frac{\sqrt{2}}{2} R_2 \tag{5}$$

From (4) and (5) we have

$$R_2 = 2R_1$$

For $R_3$

$$R_3 = R_1 + d_{min}$$

From (4)

$$R_3 = (1 + \sqrt{2})R_1$$

Using (3) we get

$$R_4^2 = 4 - (8 + 2\sqrt{2})R_1^2$$

And finally, from (1) and (4) we have

$$PAPR = 4 - (4 + \sqrt{2})d_{min}^2 \tag{6}$$

For 16-CQAM with 8 circles we follow again the algorithm which is presented at [1]. We place the 2 symbols of the first circle ($s_{11}$, $s_{12}$) on x-axis and the 2 symbols of the second circle on y-axis ($s_{21}$, $s_{22}$). Each symbol of the second circle has distance from symbols of the first circle equal to $d_{min}$. From (2) we have

$$d_{min} = 2R_1 \tag{7}$$

So for $R_2$ we get

$$2R_1 = \sqrt{R_1^2 + R_2^2}$$

$$R_2 = \sqrt{3}R_1$$

As a result, the coordinates of the above-mentioned symbols are

$$s_{11} = (-R_1, 0) \qquad s_{12} = (0, R_1) \qquad s_{21} = (\sqrt{3}R_1, 0) \qquad s_{22} = (0, -\sqrt{3}R_1)$$

We place the symbols of the third circle at distance $d_{min}$ from $s_{21}$ and $s_{22}$ without changing the y-coordinate, so for $R_3$ we get

$$R_3 = \sqrt{7}R_1$$

and

$$s_{31} = (-2R_1, \sqrt{3}R_1) \qquad s_{32} = (2R_1, -\sqrt{3}R_1)$$

We place the symbols of the forth circle at distance $d_{min}$ from $s_{11}$ and $s_{12}$ without changing the y-coordinate, so for $R_4$ we get

$$R_4 = 3R_1$$

and

$$s_{41} = (-3R_1, 0) \qquad s_{42} = (3R_1, 0)$$

We place the symbols of the fifth circle at distance $d_{min}$ from $s_{41}$ and $s_{42}$ without changing the x-coordinate, so for $R_5$ we get

$$R_5 = \sqrt{13}R_1$$

and

$$s_{51} = (-3R_1, -2R_1) \qquad s_{52} = (3R_1, 2R_1)$$

We place the symbols of the sixth circle at distance $d_{min}$ from $s_{21}$ and $s_{22}$ without changing the x-coordinate, so for $R_6$ we get

$$R_6 = (2 + \sqrt{3})R_1$$

and

$$s_{61} = (0, (2 + \sqrt{3})R_1) \qquad s_{62} = (0, -(2 + \sqrt{3})R_1)$$

And finally, we place the symbols of the seventh circle at distance $d_{min}$ from $s_{61}$ and $s_{62}$ without changing the y-coordinate, so for $R_7$ we get

$$R_7 = \sqrt{(11 + 4\sqrt{3})}R_1$$

and

$$s_{71} = (2R_1, (2 + \sqrt{3})R_1) \qquad s_{72} = (-2R_1, -(2 + \sqrt{3})R_1)$$

Using (3) we have

$$R_8^2 = 8 - (51 + 8\sqrt{3})R_1^2$$

And from (1) and (7)

$$PAPR = 8 - \left(\frac{51}{4} + 2\sqrt{3}\right) d_{min}^2 \tag{8}$$

For 16-QAM we observe that we can handle it like a CQAM with 3 circles. The first and the last circle contains 4 symbols and the inner circle contains 8 symbols. Supposing $(A_1, A_1)$ the coordinates of the $s_{11}$ we get

$$R_1 = \sqrt{2}A_1 \qquad R_2 = \sqrt{10}A_1 \qquad R_3 = R_2 \qquad R_4 = \sqrt{18}A_1$$

So for PAPR we have using (1) that

$$PAPR = 1.8 \tag{9}$$

To find the $d_{min}$ we use again (3) and we find that

$$A_1 = \sqrt{\frac{1}{10}}$$

And knowing that $d_{min} = 2A_1$ we finally get

$$d_{min} = 2\sqrt{0.1} = 0.6324555 \tag{10}$$

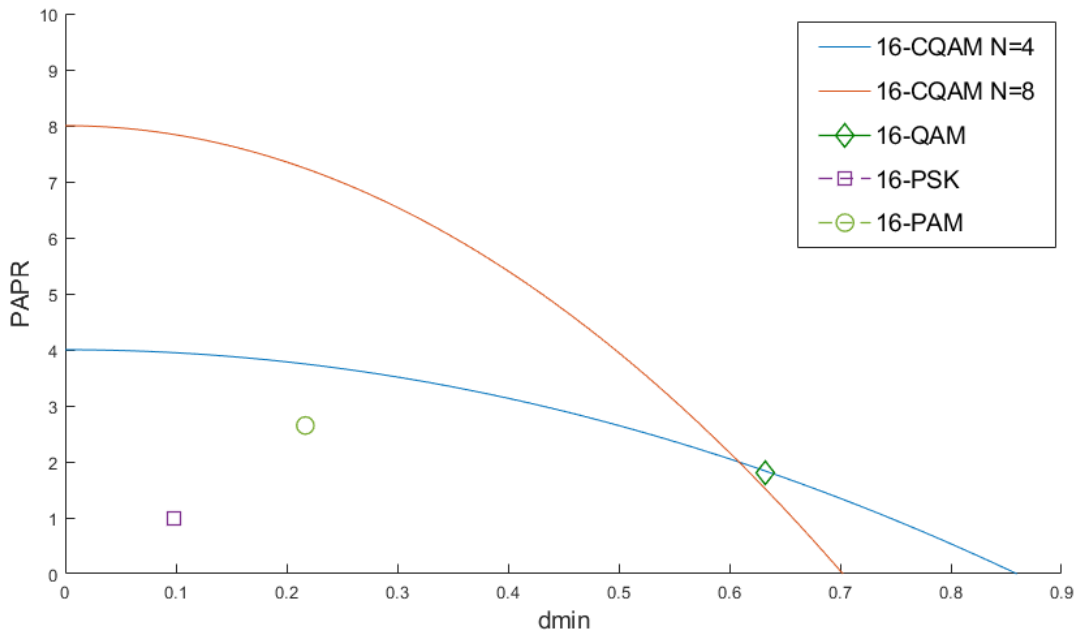


Figure 2: PAPR versus $d_{min}$ of CQAM and other famous modulation schemes

For 16-PSK we know that all the symbols are in the same circle so all of them have the same energy. For this reason we get

$$PAPR = 1 \tag{11}$$

For $d_{min}$ we have

$$d_{min} = 2Rsin\frac{\pi}{16} \approx R\frac{\pi}{8}$$

And using (3)

$$16R^2 = 1$$

$$R = \frac{1}{4}$$

So, finally

$$d_{min} = \frac{\pi}{32} = 0.09817 \tag{12}$$

For 16-PAM we observe that we can handle it like a CQAM with 2 symbols per circle. So we get

$$R_N = 15R_1$$

and

$$E_i[R_i^2] = 85R_1^2$$

From (1) we result in

$$PAPR = \frac{45}{17} = 2.647 \tag{13}$$

Using (3) we have

$$R_1 = \frac{1}{\sqrt{85}}$$

And knowing that $d_{min} = 2R_1$ we finally get

$$d_{min} = \frac{2}{\sqrt{85}} = 0.21693 \tag{14}$$

From our simulation results we achieve not only to verify the results of [1] but to find a 16-CQAM constellation with 8 circles which performs better. As a result we find out through our simulation analysis that CQAM is a more promising solution for SWIPT than the other known modulation schemes.

# 2 Task 2

## 2.1 Theoretical analysis about SER versus normalized harvested energy and SER versus SNR for 16-CQAM, 16-QAM, 16-PAM

The authors in [1] provide simulation results which illustrate SER versus normalized harvested energy with fixed SNR and SER versus SNR with fixed normalized energy harvesting comparing the performance of CQAM with the performance of QAM and PAM with 16 symbols. As we can see 16-QAM has the best performing scheme and 16-PAM provides higher PAPR, performing better at energy harvesting among these two modulation schemes. Although CQAM can give higher energy harvesting while maintaining SER at lower levels than the other two modulation schemes.
Comparing CQAM with QAM and PAM performance in terms of SER versus SNR, we notice that CQAM and QAM have almost similar results when we don't take energy harvesting into consideration and PAM gives the worst results.However, as soon as a fraction of energy of 0.2 is used for energy harvesting, CQAM clearly outperforms all existing schemes that suffer from an error floor, even 16-PAM which is known for its energy harvesting performance.
Including in our analysis and the M-ary spike QAM from [2] we notice comparing the performance of 16-CQAM with this of 16-sQAM that the sQAM outperforms CQAM as it offers more versatility, since we can perform larger values of PAPR while having the same $d_{min}$ and vice versa. Consequently sQAM performs better in terms of SER as it achieves for the same energy harvesting performance with CQAM to provide smaller values of SER.

## 2.2 Simulation analysis

According to [1], we know that the symbol error probability for CQAM is given by the equation

$$P_e \approx \frac{1}{N}\sum_{i=1}^{N} v(i) Q\left[\sqrt{\frac{E_b}{N_0} sin\left(\frac{\pi}{n}\right)\left(\sqrt{R_i^2 - \epsilon(R_i)}\right)}\right] \tag{15}$$

where

$$\epsilon(R_i) = (\rho R_i^2 - \alpha T_s P_{th}) \tag{16}$$

is the energy harvested at level $R_i$, ν(i) represents the number of neighbors at the minimum distance at level i, and Q[·] is the Q-function.

We set $\alpha = 0$ so we have $\epsilon(R_i) = \rho R_i^2$. Furthermore, we suppose that we transmit in a channel with high SNR so we set the SNR to be 20dB meaning that $\frac{E_b}{N_0} = 100$. Consequently, we have from (15) that the signal error probability for 16-CQAM with 4 circles and 4 symbols per circle,when we want the maximum PAPR, is given by the following equation

$$P_e = \frac{1}{4}\sum_{i=1}^{4} v(i) Q\left[\sqrt{50\sqrt{2} R_i \left(\sqrt{1-\rho}\right)}\right]$$

To find a relation between SEP and $\rho$ we have to set a value for $d_{min}$ in order to get a value for the radii of the CQAM. We hypothesize that $d_{min} = 0.4$, so we get

$$R_1 = \frac{0.4}{\sqrt{2}} \qquad R_2 = 0.4\sqrt{2} \qquad R_3 = (1+\sqrt{2})\frac{0.4}{\sqrt{2}} \qquad R_4 = \sqrt{PAPR} = 1.77$$

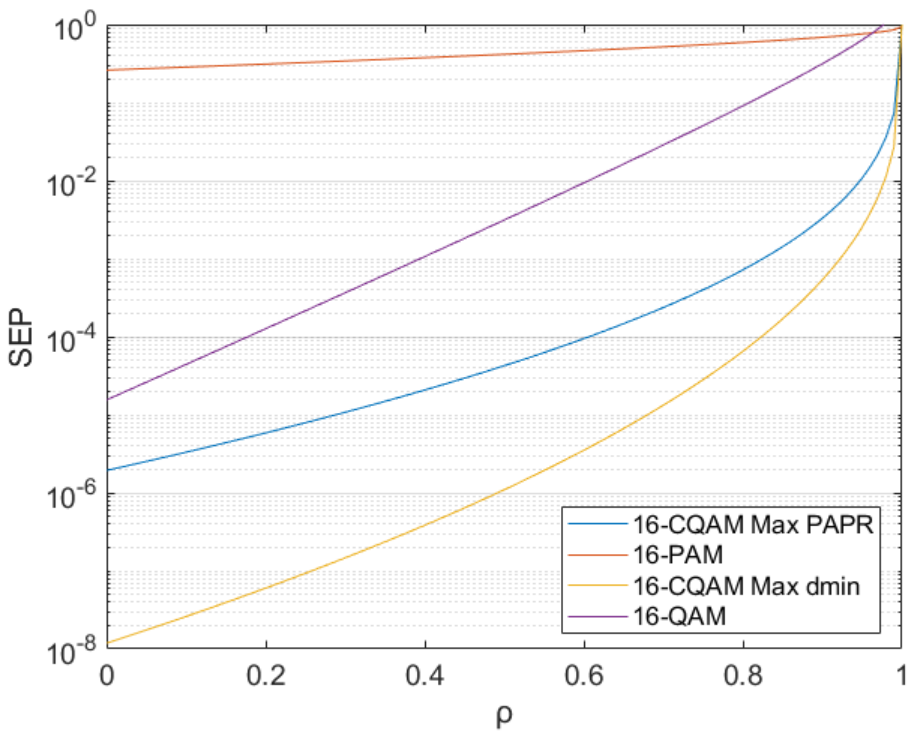


Figure 3: SEP versus $\rho$ of several modulation schemes

Hence we end up to

$$P_e = \frac{1}{4}\left(2Q\left[\sqrt{20\left(\sqrt{1-\rho}\right)}\right] + 2Q\left[\sqrt{40\left(\sqrt{1-\rho}\right)}\right] + 2Q\left[\sqrt{20(1+\sqrt{2})\left(\sqrt{1-\rho}\right)}\right] + 2Q\left[\sqrt{35.4\left(\sqrt{1-\rho}\right)}\right]\right) \quad (17)$$

where $\rho$ is a fraction of the received signal's power which is used to harvest energy and takes values from 0 to 1.

For 16-CQAM with 4 circles and 4 symbols per circle,when we want the maximum $d_{min}$ we have that

$$R_4 = R_2 + d_{min} = (2+\sqrt{2})R_1$$

We want unit average energy, so we get

$$R_1 = \frac{2}{\sqrt{14+6\sqrt{2}}} \quad (18)$$

Therefore the symbol error probability is given by the following equation

$$P_e = \frac{1}{4}\left(2Q\left[\sqrt{100R_1\left(\sqrt{1-\rho}\right)}\right] + 2Q\left[\sqrt{200R_1\left(\sqrt{1-\rho}\right)}\right] + 2Q\left[\sqrt{100(1+\sqrt{2})R_1\left(\sqrt{1-\rho}\right)}\right] + 2Q\left[\sqrt{100(2+\sqrt{2})R_1\left(\sqrt{1-\rho}\right)}\right]\right) \quad (19)$$

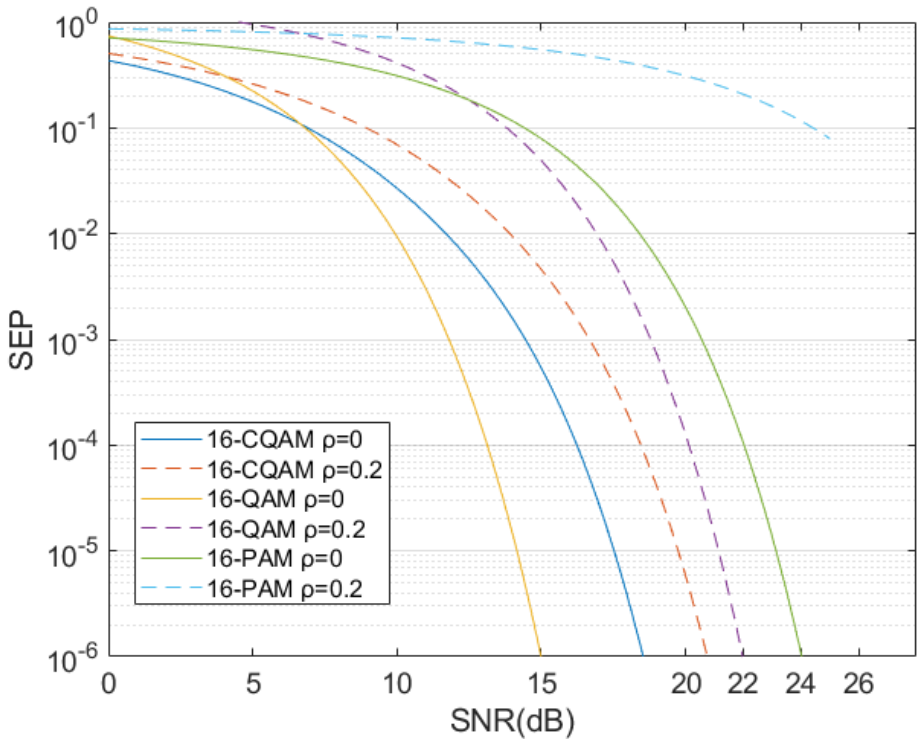


Figure 4: SEP versus SNR of several modulation schemes

For 16-QAM we measured that for unit average energy we have $R_1 = \sqrt{0.2}$ so

$$P_e \approx 4Q\left[\sqrt{100R_1^2(1-\rho)}\right] \Rightarrow$$
$$\Rightarrow P_e \approx 4Q\left[\sqrt{20(1-\rho)}\right] \tag{20}$$

For 16-PAM we know that for unit average energy we have $d_{min} = \frac{2}{\sqrt{85}}$ and $R_1 = \frac{1}{\sqrt{85}}$ so

$$P_e = \frac{15}{8}Q\left[\sqrt{100R_1^2(1-\rho)}\right] \Rightarrow$$
$$\Rightarrow P_e = \frac{15}{8}Q\left[\sqrt{\frac{20}{17}(1-\rho)}\right] \tag{21}$$

From (15) we can express the relation between SEP and SNR. For 16-CQAM with 4 circles and 4 symbols per circle, when $\rho = 0$ we have

$$P_e = \frac{1}{N}\sum_{i=1}^{N} v(i)Q\left[\sqrt{\frac{E_b}{N_0}sin\left(\frac{\pi}{n}\right)R_i}\right]$$

and we have also the maximum $d_{min}$ so replacing $R_1$ from (18) we get

$$P_e = \frac{1}{4}\left(2Q\left[\sqrt{\frac{E_b}{N_0}0.3}\right] + 2Q\left[\sqrt{\frac{E_b}{N_0}0.6}\right]\right.$$
$$\left. +2Q\left[\sqrt{\frac{E_b}{N_0}0.3(1+\sqrt{2})}\right] + 2Q\left[\sqrt{\frac{E_b}{N_0}0.3(2+\sqrt{2})\sqrt{2}}\right]\right) \tag{22}$$

When $\rho = 0.2$ and for $d_{min} = 0.4$ we have

$$P_e = \frac{1}{4}\left(2Q\left[\sqrt{\frac{E_b}{N_0}0.2\sqrt{0.8}}\right] + 2Q\left[\sqrt{\frac{E_b}{N_0}0.4\sqrt{0.8}}\right]\right.$$
$$\left. +2Q\left[\sqrt{\frac{E_b}{N_0}0.2(1+\sqrt{2})\sqrt{0.8}}\right] + 2Q\left[\sqrt{\frac{E_b}{N_0}0.885\sqrt{1.6})}\right]\right) \quad (23)$$

For 16-QAM when we don't have energy harvesting the symbol error probability is given by

$$P_e \approx 4Q\left(\sqrt{\frac{4}{5}\frac{E_b}{N_0}}\right) \quad (24)$$

And when $\rho = 0.2$

$$P_e \approx 4Q\left(\sqrt{\frac{0.8}{5}\frac{E_b}{N_0}}\right) \quad (25)$$

For 16-PAM when we don't harvest energy we have for symbol error probability

$$P_e = \frac{15}{8}Q\left(\sqrt{\frac{8}{85}\frac{E_b}{N_0}}\right) \quad (26)$$

and when $\rho = 0.2$ we have

$$P_e = \frac{15}{8}Q\left[\sqrt{\frac{0.8}{85}\frac{E_b}{N_0}}\right] \quad (27)$$

The authors at [2] have measured the equation for symbol error probability of a M-ary sQAM with i the number of symbols to be placed further apart from the center. Replacing for M = 16 we have for 16-sQAM

$$P_{sQAM} = \frac{M-i}{M}P_{\gamma QAM} + \frac{i}{M}P_{maxQAM}$$

where $P_{\gamma QAM}$ is the symbol error probability of a rectangular QAM but with a mean power smaller than 1 and $P_{maxQAM}$ is the symbol error probability of the element with the largest power against the rest of the scaled elements. For $P_{\gamma QAM}$ we have

$$P_{\gamma QAM} = \frac{3}{2}erfc\left(\sqrt{\frac{E_s}{10N_0}\gamma}\right) - \frac{9}{16}erfc^2\left(\sqrt{\frac{E_s}{10N_0}\gamma}\right)$$

where

$$\gamma = \frac{16 - iPAPR}{16 - i}$$

Taking $d_{min} = 0.4$ and i = 1 we have PAPR = 10.3 so $\gamma = 0.38$. For $P_{maxQAM}$ we have

$$P_{maxQAM} = Q\left(\frac{|\chi_M| - D_{mid}}{\sqrt{2N_0}}\right)$$

We know that $|\chi_M| = \sqrt{PAPR}$ and $D_{mid} = \sqrt{0.8}$ for a 16-sQAM, so

$$P_{maxQAM} = Q\left(\frac{\sqrt{10.3} - \sqrt{0.8}}{\sqrt{2N_0}}\right)$$

So eventually we end up to the following equation for signal error probability for a 16-sQAM with 1 symbol to be placed further apart from the center

$$P_{sQAM} = \frac{45}{32} erfc\left(\sqrt{0.038\frac{E_s}{N_0}}\right) - \frac{135}{256} erfc^2\left(\sqrt{0.038\frac{E_s}{N_0}}\right) + \frac{1}{16} Q\left(\frac{\sqrt{10.3}-\sqrt{0.8}}{\sqrt{2N_0}}\right) \quad (28)$$

From our simulation results we prove that our theoretical analysis is correct, since we verify that 16-CQAM clearly outperforms the 16-QAM and 16-PAM when it comes to symbol error probability, as with 16-CQAM we are able to harvest energy without suffering from high SEP. However we find that 16 sQAM constellation which is proposed in [2] outeperforms the 16-CQAM which is proposed in [1], whereas it can achieve higher PAPR and higher $d_{min}$ values while keeping the SEP at lower values for the same SNR.

## References


[1] G. M. Kraidy, C. Psomas, and I. Krikidis, "Fundamentals of circular qam for wireless information and power transfer," *2021 IEEE 22nd International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, pp. 616–620, 2021.

[2] M. J. L. Morales, K. Chen-Hu, and A. G. Armada, "Optimum constellation for symbol-error-rate to papr ratio minimization in swipt," *2022 IEEE 95th Vehicular Technology Conference (VTC2022-Spring)*, 2022.